Tweet Sentiment Analysis

Classifying Apple Product Tweets as Positive, Negative, or Neutral

Angela Masaki | Phase 4 NLP Project | CrowdFlower Dataset via Kaggle | *12/06/2026*

The Business Problem

Stakeholder: Apple's Marketing & Product Design Team

- Millions of tweets posted during product launches — impossible to read manually
- Negative sentiment signals product issues that need fast attention
- Teams need real-time brand health data to make informed decisions

Solution

An NLP classifier that automatically labels tweets as Positive, Negative, or Neutral — giving the team instant, scalable insight without manual review.

The Data

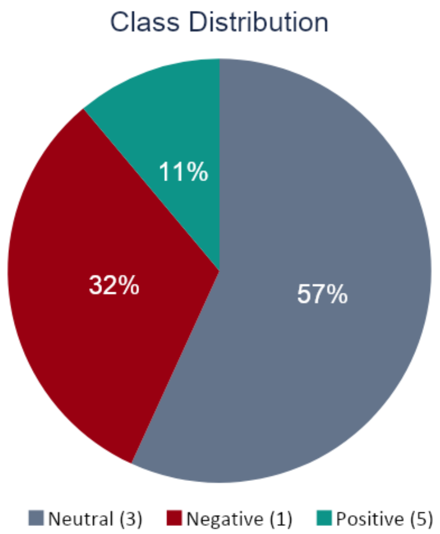
CrowdFlower Apple Twitter Sentiment Dataset

3,886
Total tweets

2014
Year collected

Kaggle
Source

Human-rated
Annotation method



Key challenge: only 11% positive tweets — class imbalance affects training

How We Prepared the Data

1	**Map Labels**	Converted numeric codes: 1 → Negative, 3 → Neutral, 5 → Positive
2	**Clean Text**	Removed URLs, @mentions, hashtags, punctuation, and numbers
3	**Remove Stopwords**	Filtered common English words (the, and, is…) using NLTK
4	**Lemmatize**	Reduced words to base form — running → run — using NLTK
5	**Feature Engineering**	Extracted tweet length, exclamation count, caps ratio, emoji proxy from raw text
6	**Vectorize**	Converted text to numbers: CountVectorizer (baseline) then TF-IDF

Libraries used: **re** (text cleaning) **nltk** (stopwords, lemmatization) **scikit-learn** (CountVectorizer, TfidfVectorizer, pipelines, FeatureUnion)

Modeling Approach

Four models, built iteratively from simple to complex:

01

Naive Bayes

Baseline — CountVectorizer, fast and simple

02

Logistic Regression

TF-IDF features, balanced class weights

03

LinearSVC

Strong text classifier, TF-IDF with bigrams

04

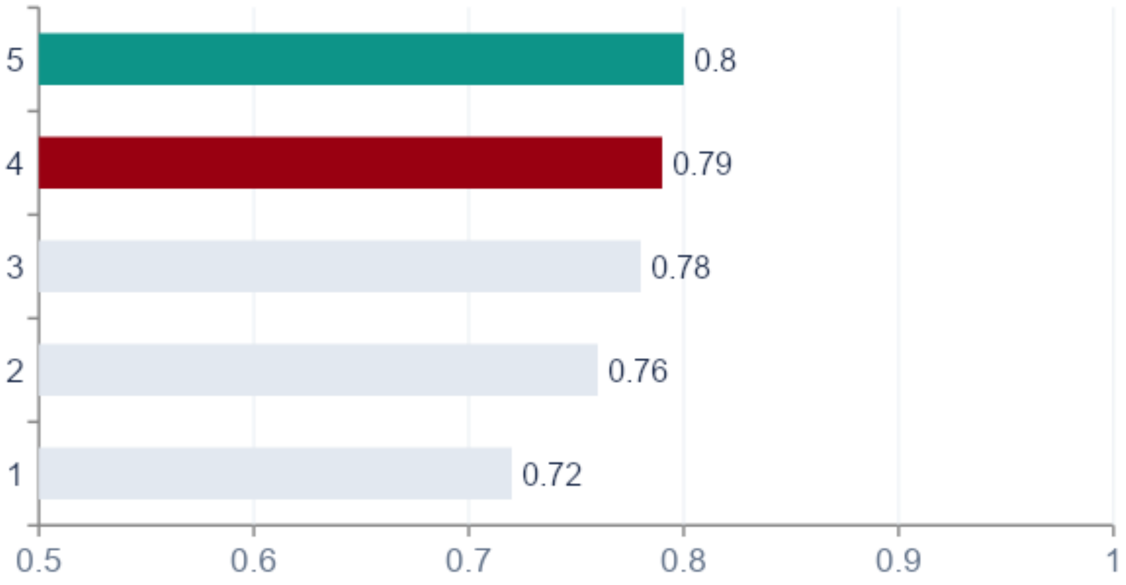
Tuned LinearSVC

GridSearchCV — optimal regularisation + vocab size

Validation: Stratified 80/20 train-test split + 5-fold cross-validation for tuning — test set never seen during training

Results

Macro F1-Score by Model — Binary Task (Positive vs Negative):



0.80
Binary Macro F1
(TF-IDF + Features)

0.79
Binary Macro F1
(Tuned LinearSVC)

0.67
Multiclass Macro F1
(3-class task)

Final model: TF-IDF + Hand-crafted Features

What the Model Learned

Top words driving each sentiment class (LinearSVC coefficients):

POSITIVE	NEGATIVE	NEUTRAL
love	fail	apple
great	sucks	iphone
best	broken	aapl
amazing	worst	stock
awesome	hate	store
nice	slow	new
good	annoying	release
excellent	bad	today
fantastic	awful	said
happy	crash	price

Limitations & Next Steps

Limitations

Only 11% of tweets are positive — minority class harder to classify

Tweets from 2014 — language and product names have evolved

Apple-only dataset — model needs retraining for other brands

No user or temporal context used in modeling

Next Steps

- Collect more positive-class tweets to balance the dataset
- Test transformer models (BERTweet) for improved short-text performance
- Add sentiment:confidence as a feature weight during training
- Deploy binary classifier as a real-time launch alert tool

Key Takeaways

Automated sentiment classification is achievable with high confidence

Macro F1 of 0.79 on binary task — ready for real-world testing

Model surfaces actionable vocabulary for product and campaign teams

A clear path to production exists — deploy as a launch alert tool

github.com/MoonwaMasaki/twitter-sentiment-nlp

THANK YOU !!

QUESTIONS?